

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<tool_call> reasoning process here

The de Broglie wavelength of a particle is given by the formula :

$\lambda = h / p$

where :

- h = Planck's constant = 6.626×10^{-34} J $\cdot$ s (or 6.626×10^{-27} erg $\cdot$ s)
- p = momentum of the particle

For a non-relativistic electron , kinetic energy K is related to momentum by :

$K = p^2 / (2m)$

where m is the electron mass = 9.109×10^{-31} kg

So , $p = \sqrt{2mK}$

Thus , $\lambda = h / \sqrt{2mK}$

We are given $K = 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

Step1 : Compute p
 $m = 9.109 \times 10^{-31} \text{ kg}$
 $K = 1.602 \times 10^{-19} \text{ J}$
 $p = \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}$
 $= \sqrt{2 \times 9.109 \times 1.602 \times 10^{-50}}$
 $= \sqrt{2.921 \times 10^{-49}}$
 $= 5.405 \times 10^{-25} \text{ kg}\cdot\text{m/s}$

Step2 : Compute $\lambda = h / p$
 $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
 $\lambda = (6.626 \times 10^{-34}) / (5.405 \times 10^{-25})$
 $= 1.227 \times 10^{-9} \text{ m}$

Convert to cm :
 $1 \text{ m} = 100 \text{ cm}$ $\implies 1.227 \times 10^{-9} \text{ m} = 1.227 \times 10^{-7} \text{ cm}$

So $\lambda \approx 1.23 \times 10^{-7} \text{ cm}$

Now look through options :

- A. 2.48×10^{-8} cm $\implies$ too small
- B. 5.92×10^7 cm/sec $\implies$ units are velocity , not wavelength
- C. 8.66×10^{-6} cm $\implies$ too large
- D. 3.14×10^{-5} cm $\implies$ too large
- E. 9.81×10^{-4} cm $\implies$ too large
- F. 5.27×10^{-9} m $\implies 5.27 \times 10^{-11}$ cm $\implies$ too small
- G. 1.23×10^{-7} cm $\implies$ Matches our calculation !
- H. 4.14×10^{-10} m $\implies 4.14 \times 10^{-8}$ cm $\implies$ too small
- I. 6.625×10^{-27} erg $\cdot$ s $\implies$ units wrong
- J. 0.512×10^6 eV $\implies$ units wrong

So the correct answer is G : $1.23 \times 10^{-7} \text{ cm}$
Note : This is correctly equivalent to $1.23 \times 10^{-9} \text{ m}$ (since $10^{-7} \text{ cm} = 10^{-9} \text{ m}$)

Therefore , the de Broglie wavelength of an electron with 1 eV kinetic energy is $1.23 \times 10^{-7} \text{ cm}$.

<answer> G </answer>